Ogbonna Arinze

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SUMMARY

Senior Fullstack Engineer with 8 years of professional experience building scalable, AI-driven web platforms across healthcare, fintech, civic tech, and enterprise environments. Expert in React and modern frontend architecture with deep backend capabilities in NodeJS, NestJS, PostgreSQL, and RESTful API design. Proven track record of shipping production RAG/LLM systems, multi-agent AI pipelines, and real-time collaborative features, with hands-on experience in Docker, CI/CD, and AWS infrastructure. Thrives in small, fast-moving teams with full product ownership and a startup mentality.

CORE COMPETENCIES

- **Fullstack Development:** React, Next.js, TypeScript, NestJS, NodeJS, Vue.js, REST APIs, and scalable component architectures.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), embeddings, multi-agent workflows, and real-time AI content generation.
- **Backend & Data:** NestJS, NodeJS REST APIs, PostgreSQL (pgvector), MongoDB, Prisma, JWT authentication, RBAC, and SSE streaming.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker, Docker Compose, GitHub Actions, CI/CD pipelines, Nx monorepo orchestration, and Traefik reverse proxy.
- **Frontend Architecture:** Design systems, monorepos (Nx), SSR/SSG, Tailwind CSS, Radix UI, Redux, Zustand, and performance optimization.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA.
- **Programming Languages:** TypeScript, modern JavaScript (ESNext), NodeJS, and Python.
- **Leadership:** Technical architecture reviews, mentorship, Agile/Scrum, TDD/BDD, and product-minded development in cross-functional teams.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer *May 2023 - Present*

- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources, embed the widget on external sites, and query their data in real time via a conversational interface.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, optimizing workflows for clinical site integrations and enhancing adoption across teams.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Implemented **JWT-based authentication** and **role-based access control** in Next.js with TypeScript, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria

Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React, Redux, RxJS**, and **Axios** for asynchronous API communication.
- Built a **real-time account reconciliation** solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic **data visualization dashboards** using React and **react-chartjs** with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)

Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria

Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng

Senior Engineer *Feb 2026 - Present*

AI-powered platform for Nigerian budget analysis and corruption tracking. Multi-agent LLM pipeline analyzes 700+ budget documents (708K+ vector embeddings) across 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app (22 interactive chart types with **Recharts**), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with **pnpm** workspaces.
- Designed and implemented a **multi-agent AI system** using the **Mastra** framework with 5 specialized agents (Router, Budget Analyst, Corruption Analyst, GovSpend Analyst, Impact Analyst) and a three-stage intent classification pipeline (keyword scoring, LLM classification with 5-min cache, inferred fallback) for domain-specific **RAG-powered** analysis.
- Built a **hybrid RAG pipeline** combining **pgvector** similarity search with BM25 full-text search, merged via Reciprocal Rank Fusion (K=60), with **Cohere** rerank-2 reranking and **Voyage AI** voyage-3-large embeddings (1024-dim, 708K+ chunks across 4 vector indexes).
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, token-budgeted context management (summary + recent messages + user profile), conversation summarization, and memory extraction.
- Built the document ingestion pipeline supporting PDF (with OCR via Tesseract), XLSX/XLS, DOCX, JSON, and Markdown extractors, processing 700+ documents (959 files) across 37 Nigerian states with zero ingestion errors.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and wrote custom migration tooling that filters out Mastra-managed chunk tables from Prisma's drift detection.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** end-to-end tests (4 projects: web auth, dashboard auth, desktop Chrome, mobile Pixel 5), and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown.
- Integrated **Langfuse** for LLM observability and scoring, **Infisical** for secrets management across environments, and **Traefik** for reverse proxy routing.

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*